RESEARCH ARTICLE

Agricultural Production Planning under Correlated Price Risk: Exact Reference Computation and Surrogate-Guided Continuous Refinement

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ARTICLE HISTORY
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ABSTRACT
Agricultural production planning couples discrete crop calendars with continuous land allocation under uncertain output prices. This study formulates a bi-objective mixed-integer planning model that maximizes expected income and minimizes covariance-based price risk while enforcing empirical crop eligibility, lot occupancy, maturation, and botanical-family recovery. Four publication-complete instances cover one or two farms, 2–8 lots, 5–12 crops, 52–104 planning weeks, and 0.6–6.0 ha. Exact ε-constraint runs in SCIP provide an incumbent reference, standard NSGA-II supplies a low-cost baseline, and two matheuristics apply either full or surrogate-guided continuous quadratic area refinement with SLSQP. Each stochastic method is evaluated over ten seeds using normalized hypervolume, IGD^+, additive epsilon, nondominated-set size, runtime, and refinement effort. Exact-run time rises from 51.25 s on the two-lot instance to 9.52 h on the largest completed instance, while most intermediate SCIP points on the larger instances terminate at the time limit. Surrogate gating reduces refinement calls from 12,000 to 3,000 per replication. On the 8-lot, 104-week instance, it reduces mean total runtime from 1,478.48 to 410.52 s relative to full refinement, but median hypervolume also declines from 0.745 to 0.718. Standard NSGA-II records the highest median hypervolume among the stochastic methods on three of four instances. The evidence therefore supports method selection as a trade-off among reference quality, computation, and planning scale, rather than a universal algorithm ranking.

KEYWORDS
agricultural production planning; multiobjective optimization; NSGA-II; matheuristics; surrogate-assisted optimization; price risk

1. Introduction

Production planning must frequently commit heterogeneous resources before uncertain economic conditions become observable, while technological or biological lead times make many assignments expensive to reverse. Agricultural planning presents this challenge in a particularly transparent form because land, planting windows, maturation periods, recovery requirements, and correlated output prices jointly determine which production sequences remain feasible and economically defensible (Ahumada & Villalobos, 2009; Tan & Comden, 2012).

The resulting planning problem cannot be reduced to continuous resource utilization, because deliberately unoccupied capacity can emerge for fundamentally different operational reasons. Some inactivity is mandatory when calendars, maturation, setup,

crop-family recovery, or horizon boundaries prevent production, whereas other inactivity is endogenous when a planner postpones an event, leaves a slot unused, or reduces cultivated area to improve the expected-income–risk compromise. Production research has long shown that zero-production periods and inserted idle time can be economically or operationally rational rather than automatically inefficient (Jodlbauer, 2006; Kanet & Sridharan, 2000; Trevino-Martinez et al., 2022; Wagner & Whitin, 1958).

Agricultural models nevertheless add empirical and structural complications that classical lot-sizing or machine-scheduling abstractions do not represent directly. Yield evidence varies geographically, crop eligibility depends on observed municipality–crop support, harvest timing follows crop-specific maturation, and crop prices move jointly across markets. Existing studies address uncertain maturation, demand, harvest, and yield, while portfolio formulations connect crop diversification with financial or environmental exposure (Burbano-Figueroa et al., 2022; Rădulescu et al., 2014; Tan & Comden, 2012); however, implementable weekly plans must also coordinate discrete event sequences with continuous cultivated areas.

This mixed discrete–continuous structure creates a consequential methodological choice between exact optimization, evolutionary exploration, and matheuristic refinement. Exact multiobjective procedures offer status-bearing incumbent trade-offs, population-based search explores extensive nonconvex schedule spaces, and conditional mathematical refinement can improve continuous allocations after discrete crop identities and dates have been decoded (Deb et al., 2002; Hu et al., 2024; Mavrotas, 2009). Refining every offspring can nevertheless become computationally dominant, which motivates surrogate-guided allocation of optimization effort rather than surrogate replacement of the economic objectives (Jin, 2011; Liu et al., 2024).

Accordingly, this study asks three questions connecting production-planning value, computational scale, and selective mathematical improvement. RQ1 examines how rapidly exact ε-constraint reference computation becomes costly as farms, lots, crops, planning weeks, and feasible events increase jointly; RQ2 compares standard NSGA-II with exhaustive and selective refinement against status-aware SCIP incumbent archives; and RQ3 evaluates whether surrogate-guided selection reduces refinement effort while retaining defensible Pareto quality under constrained runtime budgets.

The study contributes a weekly bi-objective production-planning model that maximizes expected income and minimizes covariance-based monetary price risk under empirical eligibility and agronomic feasibility, together with a controlled comparison of four computational roles. The experiment reports certification status, Pareto quality, runtime, refinement effort, and ex-post operational diagnostics without treating utilization as an optimization objective, without equating every inactive period with inefficiency, and without extending agricultural evidence into unsupported cross-industry empirical claims.

2. Related Research

2.1. Production planning under uncertainty and non-continuous resource use

Production planning under uncertainty requires explicit trade-offs among economic performance, exposure, timing, and heterogeneous resource availability, because a single utilization measure cannot distinguish productive discipline from strategically preserved flexibility. Reviews of modern planning problems document expanding opera-

tional constraints and solution families, while hybridization research shows why structured problems increasingly combine mathematical programming with metaheuristic exploration (Battaïa & Dolgui, 2013, 2022; Boysen et al., 2022).

Classical production research also establishes that inactivity cannot be interpreted uniformly as waste. The Wagner–Whitin lot-sizing model admits zero-production periods when setup and inventory economics favour temporally concentrated production (Wagner & Whitin, 1958), while the scheduling taxonomy of Kanet & Sridharan (2000) identifies inserted idle time as a purposeful decision under several performance criteria. Integrated lot-sizing and scheduling further connects production quantities, setups, and sequencing decisions (Jodlbauer, 2006), demonstrating a structural principle that transfers across sectors even when its empirical magnitude remains application-specific.

Contemporary energy-aware scheduling reinforces that distinction because planned idle time can reduce energy cost or emissions when tariffs and job sequences vary temporally. Trevino-Martinez et al. (2022) explicitly show that strategically planned idle time need not constitute waste, while maintenance-integrated scheduling uses inserted inactivity to exploit time-of-use tariffs (An et al., 2023). These studies support a cautious conceptual claim that inactivity may be purposeful, rather than an empirical claim that manufacturing results automatically generalize to agriculture.

Agricultural production planning combines that general structural distinction with biological timing, uncertain market outcomes, spatially variable productivity, and operational decisions extending from planting through harvest. The agri-food review by Ahumada & Villalobos (2009) connects planning across these decisions, while Tan & Comden (2012) represent demand, maturation, harvest, and yield risk for annual crops. Mandatory inactivity therefore arises whenever sowing calendars, maturation, family recovery, or horizon feasibility preclude another event, whereas endogenous inactivity arises from decoded waiting choices, unused event slots, and reduced cultivated areas.

Portfolio formulations provide a complementary economic perspective because crop returns are jointly distributed, meaning diversification depends on covariance rather than independent crop-specific penalties. Rădulescu et al. (2014) integrate expected return with financial and environmental risks, whereas Burbano-Figueroa et al. (2022) apply decision analysis and portfolio theory to horticultural diversification. The present setting connects that covariance-aware perspective with weekly lot occupation, continuous cultivated area, and empirical municipality–crop eligibility, while deliberately excluding weather, yield, behavioural, and complete supply-chain uncertainty.

2.2. Agricultural multiobjective planning and matheuristic solution paradigms

Agricultural multiobjective studies increasingly connect economic performance with environmental exposure, distribution, sustainability, and interactive decision support while preserving application-specific agronomic restrictions. Recent Colombian cultivation research integrates production and distribution decisions (Vergara Rodríguez et al., 2025), enhanced NSGA-II designs incorporate fuzzy information and expert rules (Huang et al., 2025), and interactive evolutionary optimization reduces evaluation demands while supporting agricultural trade-off exploration (Kropp et al., 2026).

Matheuristics exploit complementary strengths by assigning combinatorial exploration to a heuristic while mathematical programming resolves a structured subproblem or improves selected candidate solutions. In multiobjective cellular manufacturing, Hu et al. (2024) embed mathematical optimization within evolutionary scheduling,

showing how exact structure can support rather than replace population-based search. This decomposition is especially relevant when fixing discrete identities and times converts the remaining allocation problem into a more tractable continuous program.

Production applications demonstrate several ways to connect mathematical and heuristic components, including formulation-driven neighborhoods, decomposition, route improvement, and coordinated scheduling decisions. Li et al. (2022) combine an efficient formulation with a matheuristic for unrelated parallel machines, while Qin et al. (2024) address capacitated multi-level lot sizing with substitution and backorders. Related designs improve production routing (Vadseth et al., 2023) and integrate production with distribution scheduling (Vié et al., 2025), confirming that hybrid structure can broaden computational reach across planning levels.

Industrial evidence also indicates that practical matheuristics must be judged through computational reliability and decision quality under realistic planning horizons rather than through algorithm labels alone. Seyfi & Yanıkoğlu (2026) integrate lot sizing, machine scheduling, shifts, and workforce planning for washing-machine production, obtaining scalable feasible solutions when longer horizons challenge direct optimization. These studies motivate conditional mathematical refinement, but they rarely isolate whether refining every evolutionary candidate is preferable to allocating refinement through learned utility.

2.3. Surrogate-guided optimization and research gap

Surrogate-assisted evolutionary computation replaces or prioritizes expensive evaluations using models trained on previously evaluated candidates, thereby treating computation as a resource-allocation decision. Early synthesis emphasizes that imperfect surrogates require explicit model management and continued contact with true evaluations to avoid misleading the search (Jin, 2011). Multiobjective surveys similarly identify accuracy, dimensionality, infill selection, and limited evaluation budgets as central design challenges for computationally expensive optimization (Chugh et al., 2019).

More recent reviews broaden this foundation by classifying surrogate models, algorithmic roles, applications, and unresolved questions across continuous and combinatorial expensive optimization. He et al. (2023) review surrogate-assisted evolutionary algorithms across several problem classes and applications, while Liu et al. (2024) show that expensive combinatorial settings remain substantially less developed than continuous counterparts. A recent differential-evolution review likewise stresses that real-world value depends on whether surrogate overhead remains small relative to the evaluations being rationed (Yu et al., 2025).

Current algorithmic developments improve relationship learning and local-region management for expensive multiobjective problems, but their customary goal remains prediction or replacement of expensive objective evaluations. Pierce et al. (2025) learn informative comparison relationships among candidate solutions, whereas Wang et al. (2026) adapt local-region search for challenging high-dimensional expensive multiobjective optimization problems within demanding practical settings. The present study instead predicts realized optimization gain and uses that estimate only to decide which candidates receive a true continuous-refinement call.

Across these linked streams, no reviewed study jointly provides weekly agricultural scheduling, covariance-based monetary risk, exact status-aware references, exhaustive continuous refinement, and surrogate-gated refinement within one controlled experi-

ment. This gap supports a focused contribution concerning the allocation of optimization effort, while avoiding broader claims that the underlying algorithms or agricultural constraints are individually unprecedented.

3. Materials and Data

3.1. Planning setting and productivity evidence

The decision setting represents Colombian farms divided into heterogeneous lots. A production event assigns a crop, sowing week, harvest week, and cultivated area to one lot. The source productivity table contains 17,059 municipality–crop–planting-month observations, 17 municipalities, and 40 recorded agricultural products. These counts describe the source table; the completed benchmarks use supported subsets of 5–12 crops rather than all 40 products.

Yield estimation combines municipality, crop, and planting-period evidence. Crop eligibility requires at least three supporting observations for a lot's municipality–crop pair; if no crop passes that threshold for a generated lot, the crop with greatest support is retained to avoid a structurally unusable lot. Feasible event variables are created only when the sowing calendar permits the event, harvest occurs within the horizon, the empirical eligibility rule is satisfied, and expected yield is positive.

3.2. Market prices, forecasting, and uncertainty representation

Weekly prices were constructed for Bucaramanga's Centroabastos market from DANE SIPSA wholesale workbooks for 2018–2026. The input provenance records the downloaded files and hashes. The experiment uses a 104-week price forecast generated from the processed weekly series; shorter instances use its first 52 weeks. A Ledoit–Wolf covariance estimate of crop log returns supplies a positive-semidefinite dependence matrix, with numerical symmetrization and a diagonal adjustment only if required for convexity.

The optimization is performed in Colombian pesos (COP). Publication tables and figures apply a frozen reporting rate of 3,306.86 COP/USD dated 28 September 2026. If p_{kt} is COP/kg, then $p_{kt}^{\mathrm{USD}} = p_{kt}/3306.86$. Likewise, income in million COP is converted to thousand USD by multiplication by $1000/3306.86$, and risk based on million-COP exposure is converted by $(1000/3306.86)^2$. Positive constant rescaling leaves Pareto dominance unchanged; the quadratic risk unit changes by the square of the monetary conversion factor.

No `demand_weekly.csv` file was present in the reproducible computational snapshot. Consequently, the reported experiment has no market-throughput cap and makes no use of SIPSA-A demand.

3.3. Benchmark construction and empirical preprocessing

The benchmark ladder changes farms, lots, area, crop count, and horizon under fixed instance seeds. Table 1 reports the publication-complete portion of this ladder and preserves the configured progression from small single-farm cases toward larger heterogeneous planning settings.

Only instances with all expected files were admitted: one exact run and ten seeds (2026–2035) for each stochastic method. The snapshot scanner reads each

`metadata.json`, requires `status=ok`, and verifies the expected front before inclusion. It does not rely on a final run index. At the analysed snapshot, 4 instances meet this rule; the largest contains 8 lots.

4. Mathematical Formulation

The decision problem selects agronomically feasible crop events and their areas. Expected income is maximized while correlated price risk is minimized. The formulation is expressed in the internal monetary units used by optimization; the ex-post USD transformation above is used for manuscript reporting.

4.1. Sets, parameters, and variables

Let $\mathcal{F}$ be the farms, $\mathcal{L}$ the lots, and $\mathcal{L}_f \subseteq \mathcal{L}$ the lots belonging to farm f. Let $\mathcal{K}$ denote crops, $\mathcal{T}$ weekly planning periods, and $\mathcal{T}_k \subseteq \mathcal{T}$ feasible harvest periods for crop k. For an optional market extension, $\mathcal{G}$ denotes sales groups. The sets $\mathcal{C}_{lw}$ and $\mathcal{C}^q_{lw}$ contain, respectively, crop–harvest events that occupy lot l during week w and events from botanical family q whose occupation or recovery intervals conflict in that week.

Parameter A_l is lot area (m^2); n_k is crop maturation duration (weeks); $s(k,t) = t - n_k$ is the sowing week implied by harvest t; r_{lks} is expected yield (kg/m^2); and p_{kt} is expected price (COP/kg). The indicator b_{lk} equals one when empirical evidence makes crop k eligible for lot l. Matrix $\Sigma \succeq 0$ is the covariance matrix of crop price log returns, $q(k)$ is botanical family, and $g(k)$ is sales group. The optional parameter D_{gt} would cap group-g throughput in week t, but is absent from the reported base experiment.

For every feasible event (l,k,t), define

$$y_{lkt} = \begin{cases} 1, & \text{if crop } k \text{ is activated on lot } l \text{ for harvest in week } t, \\ 0, & \text{otherwise,} \end{cases} \tag{1}$$

and let $z_{lkt} \geq 0$ be cultivated area (m^2). Aggregate harvested quantity is $x_{kt} \geq 0$ (kg). Monetary exposure is $e_{kt} \geq 0$, measured in million COP, and $\rho \geq 0$ is the epigraph variable used for quadratic risk in the exact formulation.

Table 1. Publication-complete benchmark instances.

Instance	Farms	Lots	Area (ha)	Crops	Weeks
I01_micro	1	2	0.6	5	52
I02_small	1	4	1.8	8	52
I03_diverse	1	8	4.5	12	104
I04_two_farms	2	8	6.0	8	52

4.2. Bi-objective production-planning model

Production and exposure are aggregated through

$$x_{kt} = \sum_{l \in \mathcal{L}} r_{lk,s(k,t)} z_{lkt}, \qquad k \in \mathcal{K},\ t \in \mathcal{T}_k, \tag{2}$$

$$e_{kt} = \frac{p_{kt} x_{kt}}{10^6}, \qquad k \in \mathcal{K},\ t \in \mathcal{T}. \tag{3}$$

The two objectives are

$$\max \quad F_1 = \sum_{k \in \mathcal{K}} \sum_{t \in \mathcal{T}} e_{kt}, \tag{4}$$

$$\min \quad F_2 = \sum_{t \in \mathcal{T}} \mathbf{e}_t^{\mathsf{T}} \Sigma \mathbf{e}_t, \tag{5}$$

where F_1 is expected income in million COP and F_2 is correlated price risk in (million COP)^2. The latter is a monetary mean–variance measure, not a production cost.

Neither objective rewards continuous land occupation, so reduced productive intensity can become an endogenous component of the expected-income–risk compromise rather than evidence of algorithmic failure. The feasible zero-production schedule has zero monetary price exposure, while intermediate Pareto solutions can reduce exposure through smaller cultivated areas, fewer events, or longer discretionary intervals; operational use is therefore evaluated only after optimization.

Activation uses the physical lot area rather than an arbitrary large constant:

$$0 \le z_{lkt} \le A_l y_{lkt}, \qquad l \in \mathcal{L},\ k \in \mathcal{K},\ t \in \mathcal{T}_k. \tag{6}$$

Weekly conflict cliques enforce exclusive lot occupation,

$$\sum_{(k,t) \in \mathcal{C}_{lw}} y_{lkt} \le 1, \qquad l \in \mathcal{L},\ w \in \mathcal{T}, \tag{7}$$

and family-specific cliques extend this restriction through the configured recovery interval,

$$\sum_{(k,t) \in \mathcal{C}^q_{lw}} y_{lkt} \le 1, \qquad l \in \mathcal{L},\ w \in \mathcal{T},\ q. \tag{8}$$

Variables exist only if $b_{lk} = 1$, $s(k,t)$ is an allowed sowing week, t lies within the horizon, and $r_{lk,s(k,t)} > 0$. This event generation enforces empirical eligibility and temporal feasibility without adding redundant zero variables.

Optional market-throughput extension. If independent market-capacity evidence is available, the implementation can add

$$\sum_{k : g(k) = g} x_{kt} \le D_{gt}, \qquad g \in \mathcal{G},\ t \in \mathcal{T}. \tag{9}$$

Constraint (9) was inactive in every result reported here.

4.3. Epsilon-constraint reference formulation

The reference procedure first solves

$$I^{\max} = \max\{F_1 : (2) - -(8)\}. \tag{10}$$

For each configured $\eta \in \{0, 0.1, \ldots, 1\}$, SCIP then solves

$$\begin{aligned} \min \quad & \rho, \\ \text{s.t.} \quad & \rho \geq F_2, \\ & F_1 \geq \eta I^{\max}, \qquad\qquad (2) - -(8). \end{aligned} \tag{11}$$

When a feasible incumbent is found, an optional strong-Pareto tie break maximizes F_1 subject to the attained risk plus a numerical tolerance. Solver status and relative gap are retained for every point, so a feasible time-limited incumbent is not relabelled as certified optimal.

5. Solution Methods

5.1. Chromosome representation and deterministic decoder

Figure 1 summarizes the shared representation and deterministic decoding process that converts aligned crop, waiting, and cultivated-area genes into feasible dated production events.

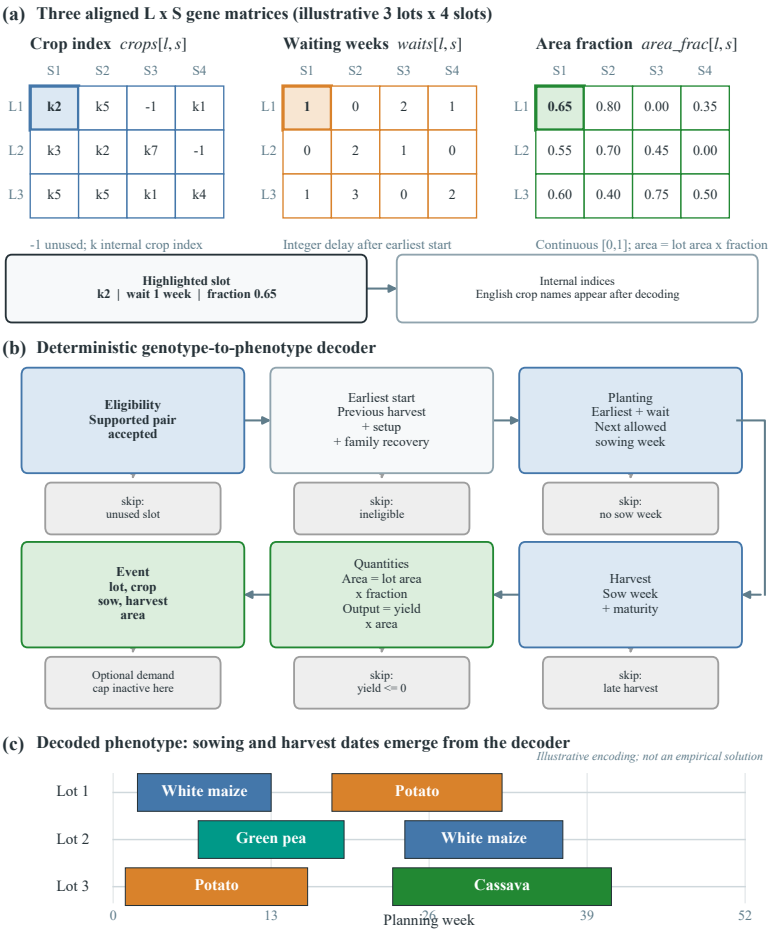


Figure 1. Chromosome semantics and deterministic decoding used by every evolutionary method. Aligned crop, waiting-week, and area-fraction matrices encode each candidate, while the decoder derives feasible sowing dates, harvest dates, cultivated areas, production quantities, and event records. The displayed schedule is illustrative rather than an empirical optimization result.

The decoder enforces empirical eligibility, temporal availability, botanical-family recovery, horizon feasibility, and positive output before evaluating income and risk. Mandatory inactivity arises from sowing calendars, maturation, setup, family recovery, and horizon truncation, whereas endogenous inactivity arises from unused slots, waiting genes permitting zero to four additional weeks, and reduced cultivated-area fractions. This distinction prevents every unoccupied interval from being interpreted automatically as scheduling inefficiency.

5.2. Common NSGA-II backbone and continuous refinement

Binary tournament selection, crossover probability 0.90, mutation, nondominated sorting, and crowding-based environmental selection operate with population 80 for 150 generations. Ten fixed seeds permit stochastic replication without changing any instance data, while every candidate retains true decoded or mathematically refined economic objective values.

Baseline NSGA-II. Standard NSGA-II retains the decoded cultivated areas and objective values without invoking continuous mathematical refinement, thereby providing the common low-cost evolutionary reference for both hybrid variants.

Full refinement.

Conditional on a decoded discrete schedule, area allocation has a linear income expression, a positive-semidefinite quadratic risk expression, physical lot constraints, and fixed event timing. The reported NSGA-II+refinement experiment sends every offspring to this continuous subproblem. Although its mathematical structure is a convex quadratic refinement, the computational backend in this experiment is SLSQP, not SCIP.

5.3. Surrogate-guided refinement

H-NSGA-II uses the same evolutionary operators and SLSQP refinement, but limits refinement to 25% of offspring. During five warm-up generations, candidates are sampled for refinement. After at least 30 realized gains are available, an Extra Trees regressor ranks candidates; 20% of each refinement budget remains random exploration. All selection, nondominated sorting, and reported objectives use actual decoded or SLSQP-refined values. Surrogate predictions only control access to refinement.

Figure 2 isolates the principal algorithmic contrast by holding evolutionary generation and Pareto survival constant while changing only how conditional area refinement is allocated. The surrogate estimates realized refinement gain for candidate prioritization, whereas every selection decision and archived objective value remains grounded in decoded or SLSQP-refined evaluations.

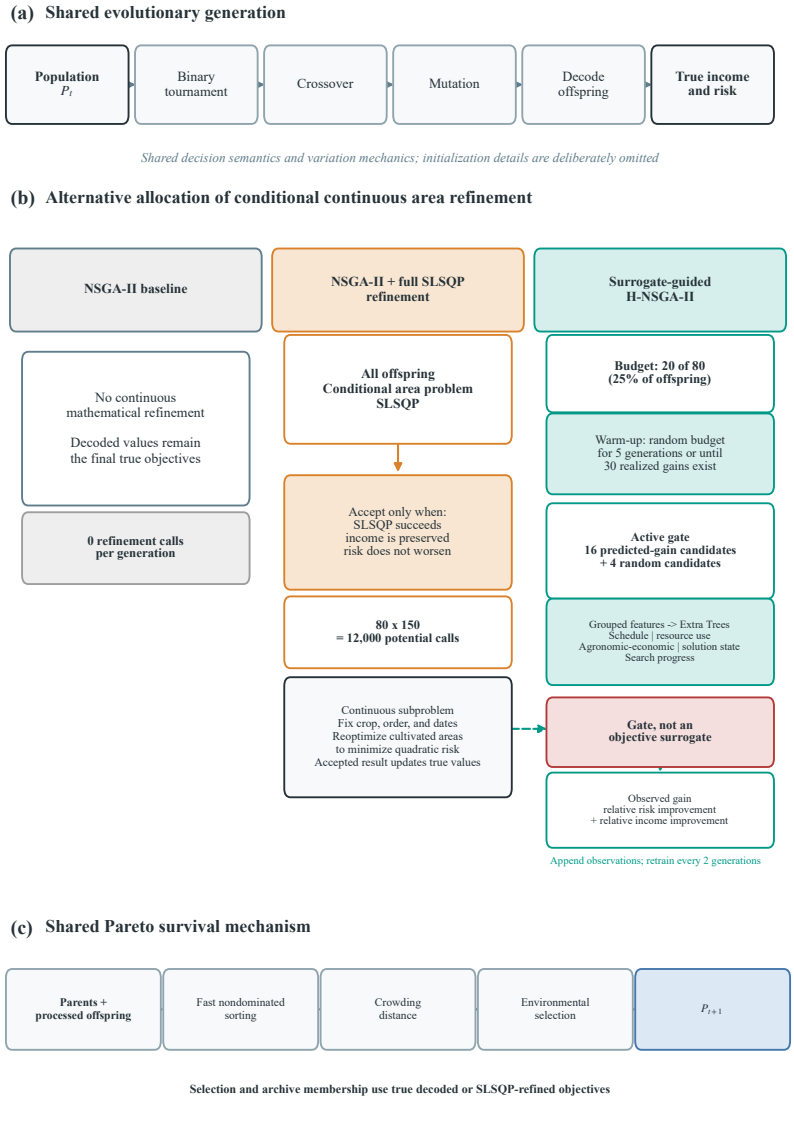


Figure 2. Shared NSGA-II backbone and alternative allocation of conditional continuous area refinement. Baseline NSGA-II performs no mathematical refinement, full refinement sends every offspring to SLSQP, and surrogate-guided H-NSGA-II allocates a 25% budget using predicted realized gain plus random exploration. All variants retain true objective values for Pareto survival.

Together, these variants isolate refinement allocation as the controlled methodological contrast, because their chromosome semantics, decoding rules, evolutionary operators, population sizes, generation counts, and objective evaluations otherwise remain equivalent.

6. Experimental Design and Evaluation

Figure 3 connects empirical inputs, configured benchmark construction, four computational methods, controlled stochastic comparisons, evaluation indicators, and the publication-completeness filter applied to the reported snapshot. This workflow distinguishes the eight configured instances from the four publication-complete instances, thereby preventing partial larger-instance outputs from entering inferential comparisons.

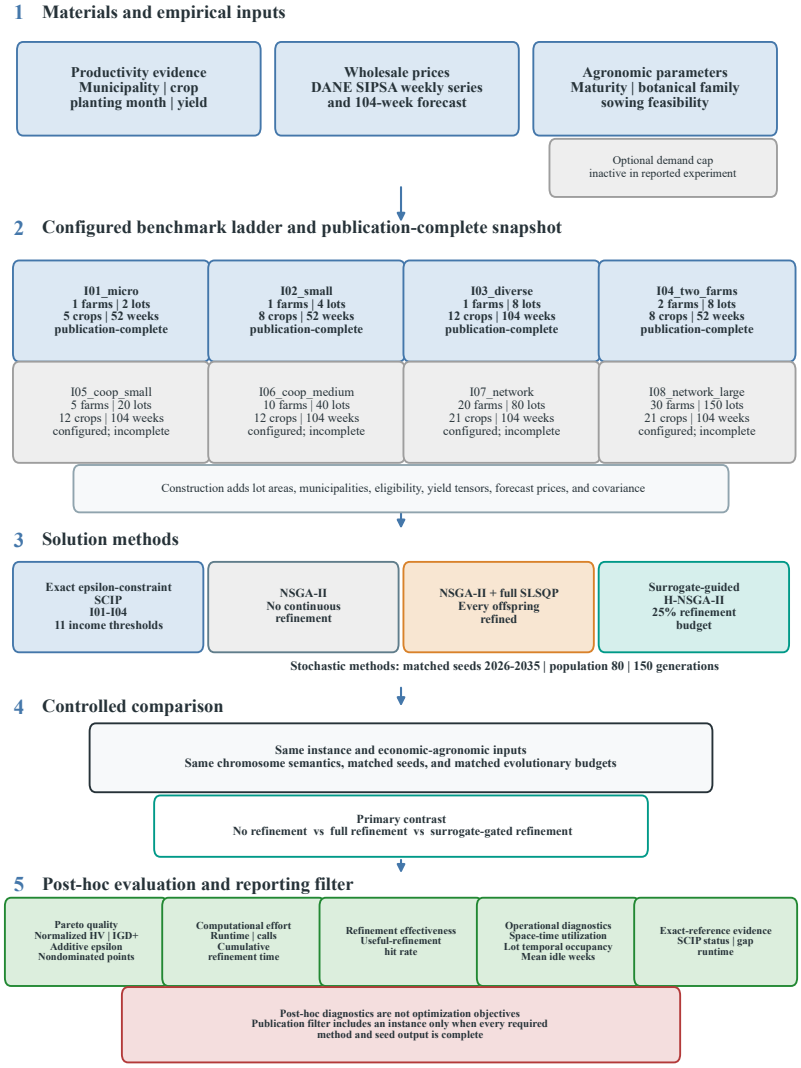


Figure 3. End-to-end controlled experiment for the publication snapshot. Materials and empirical inputs construct eight configured benchmarks; four complete instances enter exact and stochastic comparisons under matched controls, followed by post-hoc Pareto, computational, refinement, operational, and exact-reference evaluation.

6.1. Controlled computational design

All results come from `results/publication_slsqp_test`. Complete instances contain one exact run and 30 stochastic runs: ten each for NSGA-II, full refinement, and H-NSGA-II. Incomplete instances are excluded as units; a finished seed cannot enter tables unless every required seed and method for that instance is present. Optimization outputs remain unchanged, and manuscript currency conversion is reporting-only.

SCIP solves the mixed-integer quadratically constrained formulation at 11 income thresholds for I01–I04, using four threads and a target relative gap of 10^{-4}. Instance-specific limits are 300, 600, 1,200, and 1,800 s per solve, while the optional strong-Pareto tie break can require a second solve and therefore raise total runtime beyond 11 times the nominal point limit.

6.2. Pareto, computational, and operational performance indicators

For I01–I04, the nondominated SCIP incumbent archive defines the normalization and reference set. This label is deliberate because time-limited points may have nonzero or uninformative gaps. Hypervolume uses reference point $(1.1, 1.1)$ after objective normalization; higher values indicate greater dominated normalized area. Lower IGD^{+} and additive epsilon are preferred.

Nondominated point count, total runtime, refinement calls, cumulative refinement time, and useful-refinement hit rate complement these quality measures. Medians summarize stochastic quality and runtime, while mean runtime is used only for explicitly stated runtime ratios.

Operational diagnostics are calculated ex post from decoded schedules and are neither optimization objectives nor binding constraints. For a decoded event e with cultivated area a_e, sowing week s_e, and harvest week h_e, area-adjusted space–time utilization is

$$U = \frac{\sum_e a_e(h_e - s_e)}{|\mathcal{T}| \sum_{l \in \mathcal{L}} A_l}. \tag{12}$$

Lot temporal occupancy is $U_l^{\mathrm{time}} = |\cup_{e \in \mathcal{E}_l} [s_e, h_e)|/|\mathcal{T}|$, while mean idle weeks averages the horizon weeks outside those occupied unions across lots. Together, these post-hoc measures distinguish partial-area intensity from temporal occupation, although they cannot by themselves classify every inactive interval as mandatory or economically chosen.

7. Results

7.1. Exact computation and scalability

Total exact-run time increases from 51.25 s for I01 to 9,185.10 s (2.55 h) for I02, 22,824.80 s (6.34 h) for I03, and 34,274.58 s (9.52 h) for I04. Status evidence changes with scale. Ten of 11 I01 points are optimal and one terminates at the target gap. I02 has four optimal and seven time-limited points. I03 and I04 each have one optimal income-endpoint and ten time-limited points. Thus, SCIP remains valuable as an incumbent reference, but the experiment does not establish full-front optimality beyond the smallest instance.

Table 2. SCIP epsilon-constraint run and point status.

Instance	Runtime (s)	Points	Optimal	Gap limit	Time limit	Max gap
I01_micro	51.25	11	10	1	0	9.85e-06
I02_small	9185.10	11	4	0	7	1.773
I03_diverse	2.28e+04	11	1	0	10	2.376
I04_two_farms	3.43e+04	11	1	0	10	1.394

NSGA-II median runtime stays between 48.93 and 72.84 s across completed instances, whereas exact runtime grows by nearly three orders of magnitude. Full refinement raises median runtime to 1,454.71 s on I03 and 372.73 s on I04. H-NSGA-II reduces those medians to 416.35 and 187.86 s, respectively. Using means, full refinement is 3.60 times slower than H-NSGA-II on I03 and 1.97 times slower on I04.

7.2. Pareto quality

Table 3 reports seed medians. Standard NSGA-II has the highest stochastic median hypervolume on I01 (0.863), I03 (0.831), and I04 (0.831). Full refinement is marginally highest on I02 (1.009 versus 1.007 for NSGA-II), with slightly lower median IGD^+ (0.030 versus 0.032). These differences are descriptive, not evidence of a universal ordering. On I03, full and selective refinement attain lower median hypervolume than NSGA-II, while on I04 full refinement is close to NSGA-II and H-NSGA-II is lower. Conditional area optimization therefore does not translate automatically into a better final archive under fixed evolutionary settings.

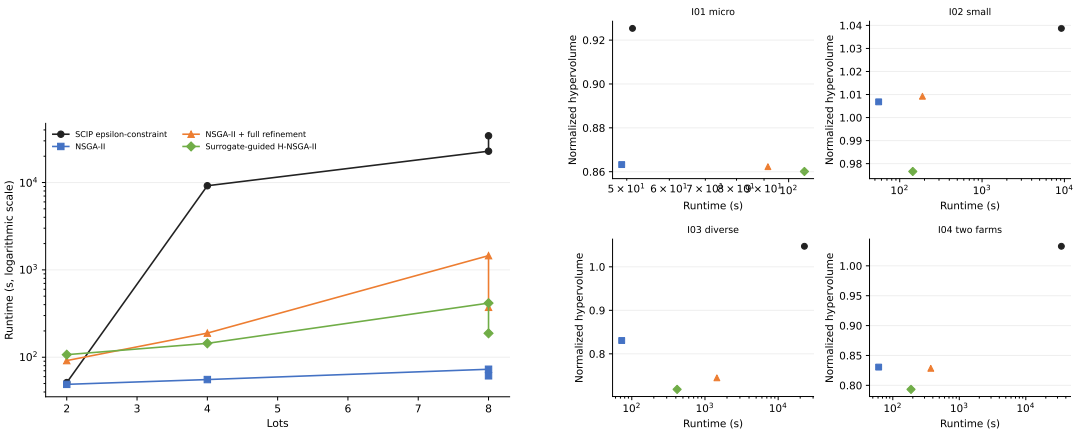
Figure 5 shows representative seeds chosen nearest each method's median hypervolume. The SCIP curves are incumbent archives; points with time-limit status must not be read as proofs of global efficiency.

7.3. Refinement efficiency and decision interpretation

Full refinement makes 12,000 SLSQP calls per replication; surrogate gating makes 3,000, a 75% reduction. On I03, median cumulative refinement time falls from 1,346.04 to 284.46 s and useful-refinement hit rate rises from 96.0% to 98.8%. On I04, refinement time falls from 289.67 to 71.54 s and hit rate rises from 94.8% to 98.9%. The

Table 3. Median Pareto quality and runtime; stochastic methods use ten seeds.

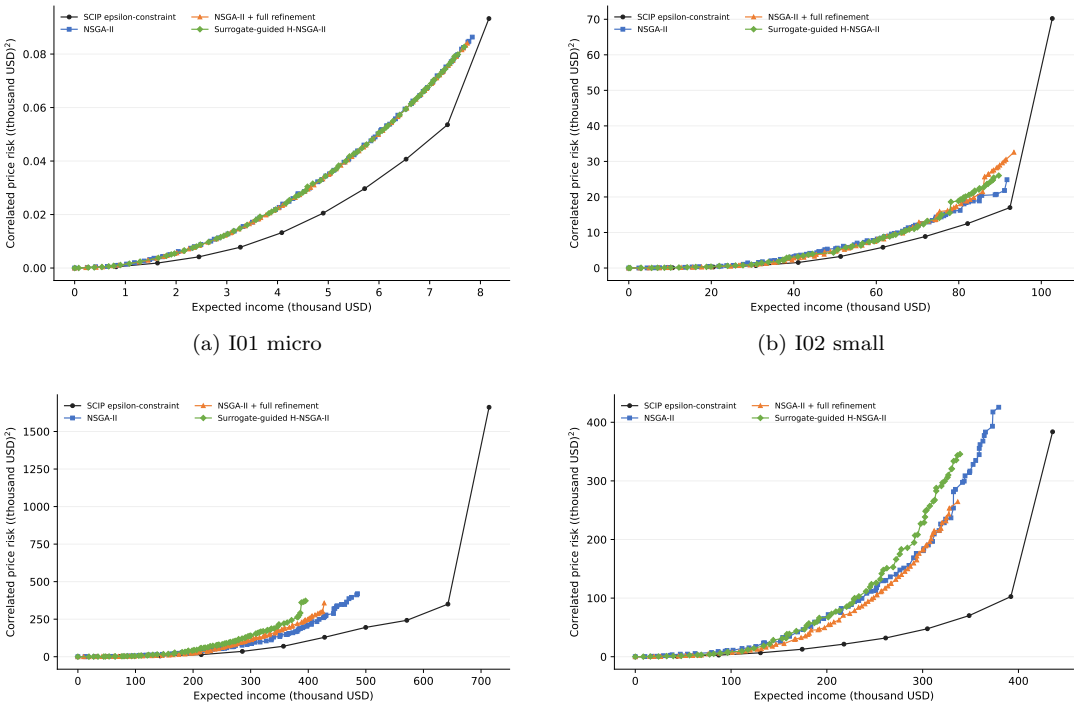
Instance	Method	HV	IGD^+	ϵ^+	ND points	Runtime (s)
I01_micro	SCIP	0.925	0.000	0.000	11.0	51.25
I01_micro	NSGA-II	0.863	0.062	0.089	78.0	48.93
I01_micro	NSGA-II + full	0.862	0.063	0.090	78.0	91.51
I01_micro	H-NSGA-II	0.860	0.063	0.089	78.0	106.98
I02_small	SCIP	1.039	0.000	0.000	11.0	9185.10
I02_small	NSGA-II	1.007	0.032	0.109	79.0	55.51
I02_small	NSGA-II + full	1.009	0.030	0.109	78.5	188.85
I02_small	H-NSGA-II	0.977	0.040	0.138	78.5	144.15
I03_diverse	SCIP	1.047	0.000	0.000	11.0	2.28e+04
I03_diverse	NSGA-II	0.831	0.081	0.306	79.5	72.84
I03_diverse	NSGA-II + full	0.745	0.104	0.400	80.0	1454.71
I03_diverse	H-NSGA-II	0.718	0.121	0.421	79.5	416.35
I04_two_farms	SCIP	1.033	0.000	0.000	11.0	3.43e+04
I04_two_farms	NSGA-II	0.831	0.116	0.210	79.0	61.28
I04_two_farms	NSGA-II + full	0.829	0.112	0.222	79.0	372.73
I04_two_farms	H-NSGA-II	0.793	0.127	0.245	79.0	187.86



(a) Runtime scaling

(b) Quality versus time

Figure 4. Computational effort and normalized hypervolume across complete instances.



(a) I01 micro

(b) I02 small

(c) I03 diverse

(d) I04 two farms

Figure 5. Income–risk fronts in publication USD units. Stochastic lines show a median-hypervolume representative seed.

higher hit rate shows concentrated allocation, but quality must be read jointly: median hypervolume falls from 0.745 to 0.718 on I03 and from 0.829 to 0.793 on I04. Surrogate gating is therefore a computational economy with a measured quality cost in these instances.

A decision-level example was selected from the median-hypervolume H-NSGA-II seed for I02. Within that front, the compromise minimizes Euclidean distance to the normalized ideal of maximum income and minimum risk. The selected point yields expected income of 59.46 thousand USD, risk of 7.52 (thousand USD)^2, and area-adjusted space–time utilization of 51.2%. Figure 7 translates that point into 11 production events over four lots, with mean temporal occupancy of 69.2% and mean idle time of 16.0 weeks per lot. The difference between area-adjusted utilization and temporal occupancy reflects partial-area events, whose allocated areas range from approximately 0.03 to 0.70 ha. These diagnostics show that white space in the Gantt combines agronomic feasibility, discretionary timing, and reduced productive intensity rather than representing a single form of scheduling inefficiency.

Table 4. Full versus surrogate-guided SLSQP refinement.

Instance	Method	Calls	Refinement time (s)	Useful (%)	HV
I01_micro	NSGA-II + full	1.20e+04	29.36	62.3	0.862
I01_micro	H-NSGA-II	3000	9.69	86.2	0.860
I02_small	NSGA-II + full	1.20e+04	114.72	90.4	1.009
I02_small	H-NSGA-II	3000	33.92	98.0	0.977
I03_diverse	NSGA-II + full	1.20e+04	1346.04	96.0	0.745
I03_diverse	H-NSGA-II	3000	284.46	98.8	0.718
I04_two_farms	NSGA-II + full	1.20e+04	289.67	94.8	0.829
I04_two_farms	H-NSGA-II	3000	71.54	98.9	0.793

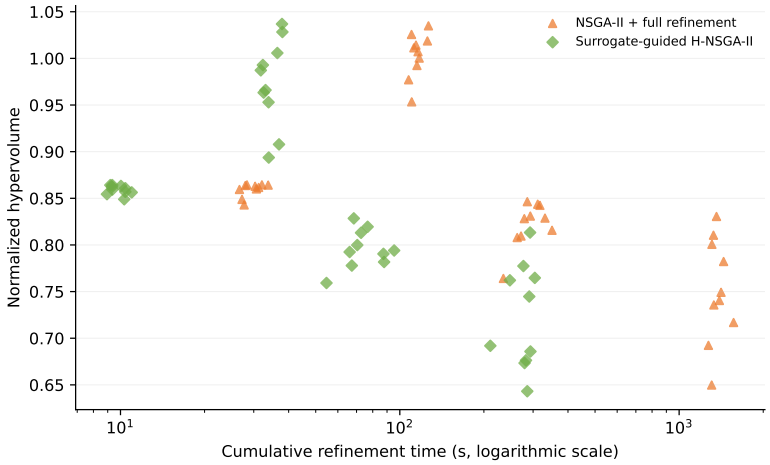


Figure 6. Normalized hypervolume against cumulative SLSQP refinement time for full and surrogate-guided refinement.

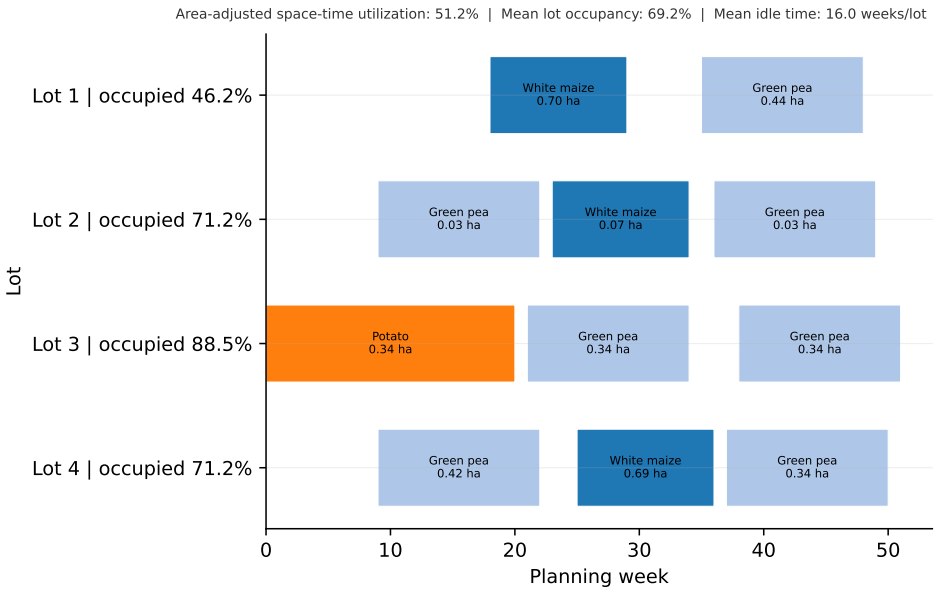


Figure 7. Representative compromise schedule for I02 small. Bars span sowing through the week preceding harvest, internal labels report crop and cultivated area, and lot labels report temporal occupancy. The upper annotation distinguishes area-adjusted space–time utilization from mean lot occupancy and reports mean inactive weeks per lot.

8. Discussion

The algorithmic comparison separates three computational roles under identical benchmark data, chromosome semantics, evolutionary operators, and stochastic seeds. SCIP supplies economically interpretable incumbent references and explicit status information, but its rapid runtime growth and time-limited points restrict certification beyond I01; standard NSGA-II supplies comparatively inexpensive diverse fronts; and continuous refinement exploits the conditional area-allocation structure without necessarily improving the global archive. Surrogate gating directs one quarter as many SLSQP calls toward candidates with higher useful-refinement rates and substantially reduces runtime on larger completed instances, although lower median hypervolume shows that omitted calls can still influence final-front quality.

The production schedules also demonstrate why inactivity should not be interpreted automatically as inefficiency. Agronomic calendars, maturation, setup, botanical-family recovery, and horizon boundaries create mandatory gaps, whereas waiting genes, unused slots, and cultivated-area reductions create endogenous gaps that may alter harvest prices or monetary exposure. Because utilization is neither an objective nor a constraint, the reported space–time utilization, lot temporal occupancy, and mean idle weeks remain post-hoc diagnostics whose interpretation requires the decoded schedule and cannot be reduced to a universal efficiency judgment.

The cross-industry literature supports a structural, rather than empirical, transferability claim. Lot-sizing models, inserted-idle-time scheduling, integrated production planning, and energy-aware scheduling all show that zero production or planned inactivity can be rational under specific costs, constraints, and temporal incentives (An et al., 2023; Jodlbauer, 2006; Kanet & Sridharan, 2000; Trevino-Martinez et al., 2022; Wagner & Whitin, 1958). Agricultural production differs in its biological calendars, spatial yield evidence, and crop-family recovery, so those manufacturing findings justify distinguishing forms of inactivity without establishing that their estimated effects or preferred policies apply directly to farms.

Several limitations bound inference from the controlled experiment. Four complete instances and ten stochastic seeds per method cannot support universal algorithm rankings, while time-limited SCIP points mean that quality indicators measure agreement with incumbent archives rather than distance to proven Pareto sets. The covariance objective captures correlated price exposure but excludes weather, yield, behavioural, and demand uncertainty; the optional market-throughput constraint remained inactive because no reproducible demand file was available; and uneven municipality–crop evidence limits empirical coverage. Future work should compare equal wall-clock budgets, vary surrogate allocation rules, complete larger benchmark ladders, and test explicitly modelled production-intensity requirements before claiming robustness under stronger utilization expectations.

9. Conclusion

This study formulates weekly agricultural production planning as a bi-objective event model combining discrete crop calendars, continuous cultivated area, empirical crop eligibility, and covariance-based monetary price risk. The formulation preserves expected income maximization and correlated price-risk minimization as its only objectives, while operational utilization remains an explicitly post-hoc diagnostic rather than an optimization target.

The computational evidence reveals a three-way trade-off among certification, Pareto quality, and response time. Exact ε-constraint computation provides status-bearing incumbent references but grows from seconds to hours and loses full-front certification as scale increases, whereas standard NSGA-II remains the lowest-cost stochastic method and achieves the highest median hypervolume among stochastic alternatives on three instances. Surrogate gating reduces refinement calls by 75% and substantially lowers refinement and total runtime on the larger completed instances, although these savings coincide with lower median Pareto quality than full refinement.

The operational interpretation is equally consequential because unoccupied time combines mandatory agronomic restrictions with endogenous economic choices, and therefore cannot be labelled inefficient without examining the decoded schedule. Larger benchmark conclusions require completion of every prescribed seed under the same snapshot rules, while future research should test stronger production-intensity requirements, richer uncertainty sources, alternative surrogate budgets, and equal-time computational comparisons before extending the present findings beyond the evaluated agricultural instances.

Data availability statement

The local reproducibility package contains source code, configuration, input provenance, benchmark metadata, result fronts, seed-level metrics, and the scripts used to regenerate every numerical table and figure. Redistribution of source data remains subject to its original provider terms.

Disclosure statement

No potential conflict of interest is reported by the author.

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